

Introductory Telecoms Theory

Engineering Technology

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Introductory Telecoms Theory (A11941)

Short Title: Introductory Telecoms Theory
Faculty Science and Computing
Department Engineering Technology
Cluster Data Communications
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Credits 5

Level: Introductory

Description of Module / Aims

This is the first telecommunications module presented in this course. Further telecommunications modules will be covered in years 3 and 4. In this module, the basic theory of analogue and digital telecommunications is introduced, from both a baseband and broadband perspective. Since it is the first of the telecommunications to be delivered, there is a greater emphasis on analogue telecommunications to establish fundamental principles and towards framing the discipline from the historical perspective. A description and analysis of different analogue and digital modulation methods are covered, modeled using both numerical techniques and software packages (e.g. Scilab) and circuit level simulators (e.g. LTSpice), and then applied to a number of historical and modern telecommunications systems. Modern transceiver architectures are explored. A practical program runs in parallel with lectures and tutorials to enhance and reinforce the student's understanding of the theory, using experience and technical information gleaned from the numerical modeling and circuit level simulations. The student is expected to have a robust grounding in mathematics, A.C & D.C. circuit theory, and analogue and digital electronics.

Programmes

	BEng (Hons) in Electrical and Automation Engineering (International) (WD_ETRIC_BI)	2 / 4 / M
TELE-0006	BEng (Hons) in Electronic Engineering (WD_EONIC_B)	2 / 3 / M
	BEng (Hons) in Information Engineering (International) (WD_EEELC_BI)	2 / 4 / M

Indicative Content

- Communication Systems Fundamentals: Modes (Simplex/Duplex) of Communication. Introduction/ motivation to/for modulation. Communications channels. Signal analysis & spectral occupancy. SNR. dB vs dBm. Upconversion/downconversion.
- Analogue Modulation: AM, FM, NBFM, PM, DSB-SC, SSB-SC (Filter, phase, Weaver).
- Basic Digital Modulation: Noise immunity. Orthogonality. ASK, FSK, BPSK, DBPSK, QPSK, 8PSK, QAM. Spectral occupancy/efficiency. Constellation representation towards intuitive error performance towards scheme/architecture like-with-like comparison.
- Communication Transceiver Architectures: Diode detector. Balanced Modulator. Varactor based oscillator. PLL recovery / demodulation. AGC. Selectivity & Sensitivity. Superhet principle / architecture. Non-coherent BPSK receiver. I&Q phase control architecture.
- Digital Telephony: RS-232. Single-ended versus balanced / differential based signalling. Sampling Theorem. TDM. PCM, DPCM/Delta Modulation. Rudimentary PABX exchange architectures. E1/T1 Interfaces.

Learning Outcomes

On successful completion of this module, a student will be able to:

1. Illustrate a good understanding of the basic theory of physical layer telecommunications.
2. Express basic physical layer telecommunications signals/architectures/operations using basic numerical modelling techniques.
3. Describe basic analogue mod/demod architectures using SPICE netlists, utilising commercial communications component netlist definitions from industry providers & use the results of the SPICE simulations to design a circuit hardware architectures.
4. Demonstrate a robust background knowledge of different types of analogue and digital communications transceivers.
5. Compare and contrast analogue and digital modulation schemes and identify a particular modulation scheme or approach with a desired deployment scenario.
6. Construct numeric & circuit level simulations of telecommunications systems, such that a partnering physical layer apparatus can be constructed in a laboratory setting.
7. Write meaningful practical reports, commensurate with a prescribed format, that includes adequate presentation of numerical / circuit simulation & experimental results, relevant calculations & meta information & draw relevant conclusions.

Learning and Teaching Methods

- Lectures
- Tutorials
- Laboratory practicals

Assessment Methods

	Weighing	Outcomes Assessed
Final Written Examination	70%	1,2,3,4,5
Continuous Assessment	30%	
<i>In-Class Assessment</i>	10%	1,2,3,4,5
<i>Practical</i>	20%	6,7

Assessment Criteria

70% -

100%: The learner has: attained the module learning outcomes at an excellent level; demonstrated a comprehensive knowledge of the associated subject matter; achieved an excellent level of the skills required for the subject matter; demonstrated the ability to carry out further investigation and problem solving associated with the subject matter.

60% - 69%: The learner has: attained the module learning outcomes at a very good level; demonstrated a detailed knowledge of the associated subject matter; achieved a very good level of the skills required for the subject matter.

50% - 59%: The learner has: attained the module learning outcomes at a good level; demonstrated a good knowledge of the associated subject matter; achieved a good level of the skills required for the subject matter.

40% - 49%: The learner has: attained the module learning outcomes at a basic level; demonstrated a basic knowledge of the associated subject matter; achieved a basic level of the skills required for the subject matter.

<40%: The learner has not: attained the module learning outcomes; demonstrated sufficient knowledge of the associated subject matter; demonstrated a sufficient level of the skills required for the subject matter.

Learning Modes

Learning Mode	F/T Hours	P/T Hours
Lecture	36	
Practical	12	
Independent Learning	87	

Essential Material

- Carlson, B. _Communication Systems_. 4th ed.. New York: McGraw Hill, 2002.
- Couch, L. _Digital & Analogue Communication Systems_. 6th ed.. New Jersey: Prentice Hall, 2001.
- Lathi, B. _Modern Digital and Analogue Communication Systems_. 3rd ed.. Oxford: Oxford University Press, USA, 1998.
- Miller, G. _Modern Electronic Communications_. 8th ed.. New Jersey: Prentice Hall, 2005.
- Tomasi, W. _Electronic Communication Systems_. 5th ed.. New Jersey: Prentice Hall, 2004.

Supplementary Material

- Dunlop, J. and D.G. Smith. _Telecommunications Engineering_. 3rd Edition. 2-6 Boundary Row, London SE1 8HN, UK: Chapman & Hall, 1994.

Requested Resources

- ENGINEERING LAB: Electronics